## Program 7: Breadth-First Search (BFS)

### Aim

To develop a Python program to implement Breadth-First Search for traversing a graph level by level.

### Algorithm

1. Represent the graph using an adjacency list.
2. Initialize a queue with the starting node.
3. Mark the starting node as visited.
4. Remove the first node from the queue and display it.
5. Add its unvisited adjacent nodes to the queue.
6. Mark the added nodes as visited.
7. Repeat until the queue becomes empty.
8. Display the traversal order.

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```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    print("\nBFS Traversal:")

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

    print()


graph = {}

n = int(input("Enter number of nodes: "))

for i in range(n):
    node = input("Enter node name: ").upper()
    neighbours = input(
        "Enter neighbours separated by spaces: "
    ).upper().split()

    graph[node] = neighbours

start = input("Enter starting node: ").upper()

if start in graph:
    bfs(graph, start)
else:
    print("Starting node not found.")
```

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```
Python 3.13.0 (tags/v3.13.0:60403a5, Oct  7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:/Users/ashik/AppData/Local/Programs/Python/Python313/7th code.py =
Enter number of nodes: 4
Enter node name: A
Enter neighbours separated by spaces: B C
Enter node name: B
Enter neighbours separated by spaces: D
Enter node name: C
Enter neighbours separated by spaces: D
Enter node name: A
Enter neighbours separated by spaces: A
Enter starting node: A

BFS Traversal:
A
```

## Result

Thus, Breadth-First Search was implemented successfully using Python to traverse the graph.